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A Robot Tour Guide for People with Sight Loss in a Robotic Assisted Living Environment

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Abstract—Robotic assisted living environments are currently lacking in accessibility for individuals with sight loss. To address this need, this preliminary study designs and evaluates a bespoke robotic tour guide which provides meaningful first-time interactions for people with sight loss in a robotic assisted living lab. We engaged 8 participants with sight loss between the ages of 20 and 80 years old, in a counterbalanced, within-subjects study comparing their experiences of a guided tour of a robotic assisted living lab led by a Temi robot, to the same guided tour led by a trained human tour guide. Results show that the robotic tour guide for people with sight loss is perceived as warm and competent, with low discomfort in the human-robot interaction. Findings also show that the tour guided by the robot is perceived to be equally engaging and useful as the tour led by the trained human tour guide, and does not lead to an increase in workload. Key findings highlight the importance of tactile exploration, multimodal interaction, and enhanced control, and identify differing needs between users who are blind and partially sighted.

Index Terms—robotic tour guide, sight loss, visual impairment, accessibility, assisted living environment

I. INTRODUCTION

More than two million people in the UK are living with sight loss (SL) [1]. Multiple research projects have aimed to use robotics to aid people with SL, mainly with navigation (e.g. [2]–[4]), using robotic guide dogs (e.g. [5]–[7]) and co-robotic canes [8], [9]. Innovative research has started to look at how social robots can be made more accessible to people with SL by focusing on meaningful first-time interactions [10], as these are important for learning functionality, shaping impressions, and building trust. However, robotic assisted living environments are currently lacking in accessibility for individuals with SL [11], [12]. To address this gap, this preliminary study introduces a robotic tour guide integrated into a robotic assisted living lab to provide meaningful first-time interactions for people with SL, and investigates the key factors influencing its effectiveness in delivering bespoke tours of a robotic assisted living environment. The robotic tour guide provides detailed auditory descriptions, facilitates interactions, and coordinates with other robots stationed along the tour route. While robotic tour guides have been implemented in various indoor and outdoor settings [13]–[15], these implementations often rely on visual cues and overlook accessibility

considerations for people with SL [10]. Even for assistive robots, standard demonstrations rely heavily on affordances perceived through visual information [10], [16], making them inaccessible to people with SL. Using a variety of interaction modalities is therefore essential for people with SL to improve clarity of communication, accessibility, acceptance, and trust [10], [17]–[20].

Human tour guides have played a crucial role in providing accessibility for individuals with SL by providing engaging, personalized experiences including verbal descriptions of the surroundings, guiding them safely through complex environments, and adapting their communication style based on their preferences and requirements [21]–[24]. However, human tour guides trained for the specific needs of people with SL are often not available, and if they are, there is variability in the delivery of information due to differences in training, experience, personal biases, or fatigue [21], [22], [25].

Robotic tour guides are a promising alternative to human guides, particularly in environments where continuous human assistance may not be feasible. Studies have shown that robotic guides can efficiently provide navigation and information in controlled environments, but often struggled in interpreting complex social cues, adapting to unexpected changes, or providing unscripted (additional) assistance [13], [26]). They also often lack in creating a personalised and engaging experience [26], which is especially important in first-time interactions with social robots [10], [27]. Robotic tour guides also offer several advantages though, such as consistency in information delivery [21], [22], increased autonomy for people with SL [28], and are more suitable in environments where privacy is a concern, such as medical facilities and assisted living environments [19], [29].

The overall aim of the research presented in this paper is to design and evaluate a robotic tour guide of a robotic assisted living environment, for people with SL, guided by the following research questions:

RQ1: What are the key factors that influence the effectiveness of a robotic tour guide for individuals with sight loss of a robotic assisted living environment?

RQ2: What is the difference, if any, in (a) perceived workload, (b) engagement, and (c) perceptions of social attributes of robots, between a human- and robot-guided tour in this context?



Fig. 1. Temi robotic tour guide, with a participant who is blind putting their right hand on the robot expecting it to physically guide them.

II. ROBOTIC ASSISTED LIVING LAB

Our robotic assisted living lab consists of a fully equipped kitchen, dining area, sitting room, two bedrooms and a bathroom (see Figure 3). It is a typical home setting and fully functioning apartment. It also is the home of various robots, such as the Temi, TIAGo, and Pepper. The apartment, which is part of a larger facility housing multiple labs for robotics and AI, is part of a guided tour for visitors from all over the world. However, these tours as well as the robotic assisted living lab are lacking in accessibility for individuals with SL, and human tour guides trained in the specific needs of individuals with SL are not available. This is a common issue across many different types of venues [25].

III. SYSTEM DEVELOPMENT AND IMPLEMENTATION

Our robotic tour guide is designed and developed to provide an accessible and engaging tour experience for people with SL of our robotic assisted living lab. In order to select the robots featuring on the tour, including the robot tour guide, we undertook a consultation with user experts from Sight Scotland, a charity supporting people with sight loss. The resulting system integrates three distinct robots—Temi (i.e. robotic tour guide), and the Tiago++ and Pepper (i.e. both socially assistive robots which will feature in the guided tour).

A. Temi: Robotic Tour Guide for People with Sight Loss

Temi (see Figure 1) was selected as the robotic tour guide for people with SL due to its compact size, suitability for moving around, and its large interactive touchscreen. The touchscreen enables touch interaction and the display of the audio description of the tour as text on the screen for people who are partially sighted (see Figure 1).

Temi was programmed to autonomously navigate the robotic assisted living lab (see Figure 3) with predefined stops in the kitchen, sitting room, bedroom, and bathroom. It provides clear audio descriptions of the environment, enhancing participants' orientation and understanding of their surroundings. A detailed map of the robotic assisted living lab was created using Temi's LIDAR (Light Detection and Ranging) system and integrated sensors, helping it to avoid static obstacles and moving participants in real-time. The user interface consists



Fig. 2. Pepper, in bedroom 1, with bright green tape for additional colour contrast for individuals with partial sight.

of a single-screen Android application with buttons enabling them to start, pause, resume, or stop the tour at any time, giving them control over the pace and flow of information.

The audio descriptions at each tour stop were pre-programmed into the application. This approach was chosen to maintain consistency in tour information. Text-to-speech (TTS) technology was utilised to voice the textual descriptions through Temi's onboard speakers.

B. Tiago++ and Pepper: Assistive and Social Interactions

Tiago++ was placed in the dining area and programmed to navigate to the sitting room where participants were seated on the sofa, and hand them a shopping bag with food. It then moves away before describing itself using its standard demo and its onboard Text-to-Speech functionality.

Pepper, a social robot (see Figure 2), acting as a social gaming companion, was located in the bedroom. The memory game played with Pepper was used in a study with older adults to help engage them in cognitive activities [30], [31], but was also found to be engaging for people with sight loss [10]. The game involves the robot asking the person to locate and touch different parts of Pepper's body in a specific order.

IV. METHODOLOGY

Participants were recruited via the charity Sight Scotland, and by distributing flyers via email and physically across the

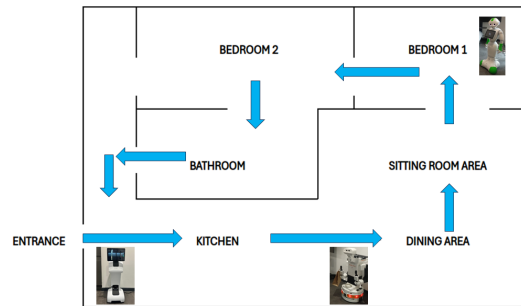


Fig. 3. Route of the guided tour through the robotic assisted living lab

TABLE I
PROTOCOL

Phase	Duration
Pre-tour Preparation	10 mins
Tour Execution (2x, counterbalanced, RoSAS per robot)	20 mins per tour
Post-Tour Evaluation (NASA-TLX+IMI per tour)	15 mins

university campus. Each participant received an Amazon gift voucher worth \$13.

A within-subjects experimental design was adopted and counterbalanced to minimise order effects and biases. Participants were randomly assigned to each condition (i.e. robot- or human-guided tour first). The tour route (see Figure 3) and tour script for the human- and robot-led tour were the same as well as the (positions of the) two robots encountered as part of the tour (i.e. TIAGo and Pepper), to ensure the same information was provided across both guided tours [32]. The human tour guide was a HRI researcher, trained by Sight Scotland to provide guided tour for people with SL. This way both human and robotic tour guides were tailored to people with SL. Two researchers and one sighted guide were present in the lab during both tours for health and safety reasons.

The study's protocol is outlined in Table I. For all robots in each guided tour, RoSAS was used to measure participants' perceptions of social attributes [33]. IMI was used to measure engagement [34], [35] and NASA-TLX to measure the perceived workload [36] of the human- and robot-guided tour. All questionnaires were verbally administered by a researcher who was not the tour guide. During each tour, participants got the opportunity for tactile exploration of each robot prior to its interactive demo, as advised by [10].

V. RESULTS

A. Participants

Eight participants (7m, 1f) aged 20-80 years old took part in this study. This included individuals with congenital blindness (P1, P6) and acquired low vision from degenerative or age-related conditions (P2, P3, P4, P5, P7, and P8). None had prior experience with any of the three robots or with a robotic assisted living environment. The study received ethical approval from the university, and a health and safety visit was conducted by a staff member of Sight Scotland.

Due to the limited number of participants, no statistical testing was done on the quantitative data, only descriptive statistics are reported.

B. RoSAS: Perception of Social Attributes Questionnaire

Participants perceived the robotic tour guide to be warm ($M=5.88$, $SD=1.35$) and competent ($M=5.44$, $SD=1.52$), and experienced little to no discomfort while interacting with the robotic tour guide ($M=1.75$, $SD=1.29$). 6 out of 8 participants rated the robotic tour guide for people with sight loss very high (i.e. ≥ 5.50) on warmth, and 5 out of 8 participants rated it very

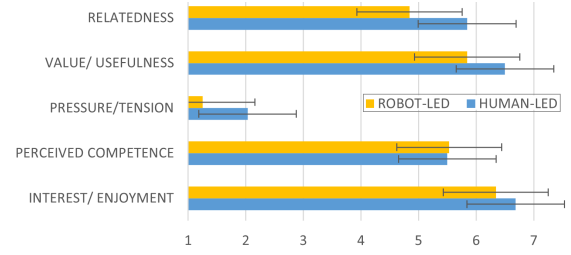


Fig. 4. Average IMI subscale scores of the robot-led tour vs human-led tour.

high competence (i.e. 6.25-7), while 7 out of 8 participants rated it very low on discomfort (i.e. 1-2).

C. IMI: Engagement Questionnaire

For engagement, the averages of each of the subscales of the IMI were calculated for each guided tour (see Figure 4). Interest and Enjoyment, Perceived Competence, and Value and Usefulness were classified as very high (i.e. ≥ 5.50) in both the human- and robot-led tour. Perceived Pressure and Tension during the human-led and robot-led tours were very low, while Relatedness to the tour guide was positive for both.

D. NASA-TLX: Workload Questionnaire

The average workload for each of the subscales was calculated (see Figure 5). Mental Demand, Physical Demand, and Effort were medium, while Temporal demand was low in the robot- and human-led tour. Perceived performance was high for both tours. Finally, Frustration was medium in the robot-led tour, but low in the human-led tour.

It should be noted though that a within-subjects t-test showed that one individual participant (P4) experienced a significantly higher overall workload in the human-led tour compared to the robot-led tour ($t(6) = -3.13$, $p=0.03$), in particular for mental-, physical-, and temporal demand, but also felt they performed better in the human-led tour.

E. Observations

In both conditions participants commented on being positive about the level of detail in the audio description of the assisted living environment. Observations of participants during the robot- and human-led tours highlighted the importance of physical exploration when interacting with these technologies in a robotic assisted living environment. Elements within the environment (e.g. a button to adjust the height of the kitchen counter) were explored using one or both hands to get an idea of where they are positioned and their size. During the robot

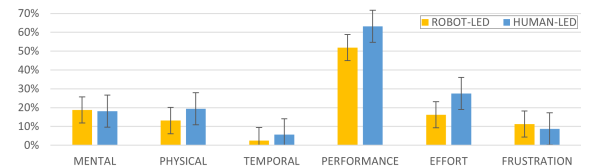


Fig. 5. Average NASA-TLX subscale scores of the robot-led tour vs human-led tour.

introductions a pattern of tactile exploration was observed. Participants started at the top of the robot and moved to the bottom, pausing at areas of interest. Sometimes they used both hands, patting down the left and right sides of the robot, but participants who used a (white) cane as a mobility aid tended to only use one hand to explore the environment and the robots. However, they would use the cane during the interaction with the Pepper to interact with its touch sensors at the base.

Observations during the robot-led tour revealed a difference in needs between participants who are blind and those who are partially sighted. Blind participants would put their hand on the Temi robotic tour guide, expecting it to physically guide them around the assisted living lab (see Figure 1). Firstly, they were unsure where to position their hand. Secondly, the robotic tour guide had not been programmed to act like a sighted guide. It moved at a speed that, especially participants with a walking aid, could not physically keep up with, resulting in them letting go of the robot. Thirdly, while trying to follow the robot, they would use their walking aid to try and track its movement or find its location. Several participants failed to do so, and would wait for the robot to start speaking again to locate it and move towards it, or had to be guided by the researcher.

Participants who were partially sighted also indicated that they sometimes struggled to visually locate (parts of) of the different robots, especially if they were stationary. This was due to the lack of colour contrast on the robots themselves, as well as against the environment, as both mainly consisted of different shades of white and grey (see Figure 1).

VI. DISCUSSION

This paper presents the design and preliminary evaluation of a bespoke robotic tour guide for individuals with SL in a robotic assisted living lab. Results in Section V-B show the robotic tour guide was perceived as warm and competent, and did not appear to influence how the social attributes of the other two robots encountered during the tour were perceived (RQ2c). IMI results in Section V-C showed interest and enjoyment, perceived competence, and value and usefulness were very high (i.e. ≥ 5.50) for both guided tours while pressure and tension were low. This been shown by [27] to be success criteria for robotic tour guides. These results suggests that the robotic-guided tour is perceived as equally engaging and useful as the human-guided tour (RQ2b). Results in Section V-D show that having a robotic tour guide, on average, did not appear to increase the workload for individuals with SL on any of the subscales (RQ2a), except for frustration, which was low ($M=8.75$) in the human-led tour and medium in the robot-led tour ($M=11.25$). One participant (P4) however had a significantly higher workload in the human-led tour, which indicates that for individual cases, there might be a preference for a robotic- or human tour guide to fit their individual needs.

Although results showed low discomfort (i.e. 1.75 out of 7) on average for the robotic tour guide and the workload across the various subscales was low to medium in both conditions, observations identified some key factors that influenced the

effectiveness of the robotic tour guide (RQ1). Unlike partially sighted participants, blind participants wanted and expected to be physically guided through the assisted living environment by the robotic tour guide. This in line with recent studies on assistive autonomous navigation robots for blind people [7], [37], [38], but appears to extend to robotic tour guides. Incorporating a physical handle for them to grab in the design of the robot could be helpful to this end. The robotic tour guide also moved too fast for blind individuals to keep up with it, and once they lost the robot, needed assistance finding its location, which may have contributed to the higher level of frustration for the robot-led tour. Since most participants also used mobility aids (meaning both hands were full while moving around), implementing a voice-controlled speed adjustment as well as an option enabling the user to find its location through voice commands, is recommended to enhance control and further accommodate users' autonomy and varying accessibility needs. The need for tactile exploration in first-time interactions with social robots for people with SL identified by [10], appears to extend to robotic assisted living environments. The importance of multimodal interaction in HRI has been highlighted before (e.g. [10], [17]). But we found this appears to be particularly important for adults with sight loss as they are likely to have additional physical disabilities [39], and rely on and try to combine all modalities (including residual vision). Therefore, we recommend to incorporate high visual contrast for key areas in both the physical design of robots (see Figure 2), as well as the robotic assisted living environments.

A limitation of this preliminary study is the small sample size ($N=8$), which also lacks diversity. Future work will focus on conducting a larger study for statistical testing and comparison, and with more diversity in gender and age. Free-response questions to obtain more qualitative insights regarding the user experience of both guided tours will be include, as well as a longitudinal study to obtain deeper insights into the evolving the needs of people with SL regarding interactions with robots in a robotic assisted living environment over time.

VII. CONCLUSION

The design and preliminary evaluation of a bespoke robotic tour guide for individuals with SL to enable positive first-time interactions in a robotic assisted living lab, show that the robotic tour guide is perceived as warm and competent, with low discomfort in the human-robot interaction. The tour guided by the robot appears to be equally engaging and useful as the tour led by the trained human tour guide, and does not lead to an increase in perceived workload. Key findings highlight the importance of tactile exploration, multimodal interaction, and enhanced control, and identify differing needs between users who are blind and partially sighted.

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